Band-1 Hadamard order 14644 – Classical Paley-II DIRECT (q2=7321); IN SPK via Paley-II; Gap 2052 Still OPEN

H_14644 Paley-II PASS; Paley-I FAIL/SKIP (q=14643=3^2*1627 NOT prime); Kronecker FAIL/SKIP (4*3661; 3661=7*523 odd NOT Hadamard-admissible); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-13

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-10-13_Band1_Hadamard_14640_PaleyI` (current bookkeeping); `2026-10-13_Band1_Hadamard_146` `2026-10-13_Band1_Hadamard_14620_PaleyII` (Paley-II construction template); `2026-10-13_Band1_Hadamard_14` (shape-class peer ===4/m===1); `2026-10-09_Band1_Hadamard_14432_PaleyI`; `2026-10-09_Band1_Hadamard_1` `2026-10-09_Band1_Hadamard_14356_PaleyII`; `2026-09-21_Band1_Hadamard_9124_PaleyII`; ....
Lab. `/workspace/experiments/band1/hadamard_14644_paleyII/`.
Cutoff. 2026-10-14T00:44:16Z (~8:44PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 14644 only (verified max|HH^T-nI|=0), via classical Paley-II q2=7321 (prime ===1 mod 4), order 2(q2+1). 14644 IN SPK via Paley-II. 14644 is a long-known Hadamard order; this is a finite on-box verification, NOT a new-order claim.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14644: Paley-II **PASS** ($q_2 = 7321$ prime $\equiv 1 \pmod 4$, $n = 2(q_2+1)$); Paley-I **FAIL/SKIP** ($q = 14643 = 3^2 \cdot 1627$ NOT prime); Kronecker **FAIL/SKIP** ($14644 = 4 \cdot 3661$; $3661 = 7 \cdot 523$ odd composite, NOT Hadamard-admissible; do NOT invent H_3661). **14644** folds into SPK **only** via classical Paley-II (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED** (no invented J). No finished-ToE claim. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14640+14628+14620+14596+14592+14564+14552+14544+14520+14508+14504+14480+14476+144** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

Abstract

Finite construction of order **14644** (eighth Paley-II DIRECT wave of the 14xxx chain, after 14356, 14428, 14460, 14476, 14508, 14596 and 14620).

- H_{14644} = Paley-II ($q_2 = 7321$) – classical SPK
- Paley-I for 14644: **FAIL/SKIP** ($q = 14643 = 3^2 \cdot 1627$ NOT prime)
- Kronecker for 14644: **FAIL/SKIP** ($4 \cdot 3661$; $3661 = 7 \cdot 523$ odd composite, NOT Hadamard-admissible)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-II.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_14644.py`

Results. overall_pass=True; units 900/900; row blocks 115/115 (block size 128); wall 6408.1739s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 14644 = 2(q_2 + 1)$ with $q_2 = 7321$ prime $\equiv 1 \pmod 4$ — Paley-II eligible (claim path).
- Paley-I: $q = n - 1 = 14643 = 3^2 \cdot 1627$ NOT prime — FAIL/SKIP.
- Kronecker: $14644 = 4 \cdot 3661$; $3661 = 7 \cdot 523$ odd composite ($\equiv 3 \pmod 4$) NOT Hadamard-admissible — FAIL/SKIP; no other admissible split.
- Shape: $14644 = 2^2 \cdot 7 \cdot 523 \equiv 4 \pmod{16}$ ($m = 3661 \equiv 1$ odd, NOT Hadamard-admissible; shape peer $14596 = 2^2 \cdot 41 \cdot 89$; template 14620 was $\equiv 12$ / $m \equiv 3$).
- Miyamoto digression skipped this wave (prefer classical SPK).

Construction (Paley type II)

Let χ be the quadratic character of $\mathbb{F}_{7321}$ and $Q_{ij} = \chi(j - i)$ (symmetric, since $q_2 \equiv 1 \pmod 4$). The conference matrix $C = \begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & Q \end{pmatrix}$ of order $q_2 + 1 = 7322$ gives

$$H_{14644} = \begin{pmatrix} C + I & C - I \\ C - I & -(C + I) \end{pmatrix}.$$

Built directly in int8 (memory-lean; unit-tested equal to the int64 reference builder on $q \in \{5, 13, 17, 29, 37, 41, 53, 61\}$), then verified block-wise: every row block of $HH^T - nI$ checked exactly in int32.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14644 Paley-II	Paley-II q2=7321	0	6387.811
H_14644 Paley-I	FAIL/SKIP (q=14643=3ˆ2*1627 NOT prime)	–	–
H_14644 Kronecker	FAIL/SKIP (4*3661; 3661=7*523 odd NOT Hadamard-admissible)	–	–

SPK closure note

Quantity	Value
14644 in SPK	**True** (Paley-II q2=7321)
Prior Paley-I 14640	IN SPK via Paley-I q=14639 (~6:45PM ET 2026-10-13)
Prior Paley-I 14628	IN SPK via Paley-I q=14627 (~4:42PM ET 2026-10-13)
Prior Paley-II 14620	IN SPK via Paley-II q2=7309 (~3:13PM ET 2026-10-13)
Prior Paley-II 14596	IN SPK via Paley-II q2=7297 (~1:12PM ET 2026-10-13)
Prior 14592	IN SPK via Paley-I (~11:06AM ET 2026-10-13)
Prior 14432	IN SPK via Paley-I (~7:51PM ET 2026-10-09)
Prior Paley-II 14428	IN SPK via Paley-II q2=7213 (~7:10PM ET 2026-10-09)
Prior Paley-II 14356	IN SPK via Paley-II q2=7177 (~1:22PM ET 2026-10-09)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false

- GATE-04 LOCKED (no invented J)
- No finished-ToE claim
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- 14644 is a long-known order (Paley 1933 family); finite on-box verification only
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

NEXT direct route after 14644 (FIRST any direct = Paley-II): q2=7333->n=14668 Paley-II DIRECT (prime ===1 mod 4; Paley-I q=14667=3*4889 FAIL; Kronecker 14668=4*3667; 3667=19*193 ===3 mod 4 NOT Hadamard-admissible -> FAIL/SKIP; template hadamard_14644_paleyII / hadamard_14620_paleyII) | next Paley-I q=14683->n=14684 (template hadamard_14640_paleyI) | Kronecker-only NOTE 14576=4*3644 (H_3644 via Paley-I q=3643; below 14644, note-only, unbuilt) | skipped Paley-II DIRECT orders below 14644 still unbuilt (template hadamard_14644_paleyII / hadamard_14620_paleyII): 13924 (q2=6961) / 13956 (q2=6977) / 13996 (q2=6997) / 14004 (q2=7001) / 14028 (q2=7013) / 14116 (q2=7057) / 14140 (q2=7069) / 14220 (q2=7109) / 14260 (q2=7129) (+13900 q2=6949 historically not listed)

Erratum (2026-10-13 ~9:00PM ET)

The first emission of this memo (~8:44PM ET) carried stale template text: wrong factorizations $14643 = 3 \cdot 5^2 \cdot 193$ and $3661 = 7 \cdot 11 \cdot 47$ (correct: $14643 = 3^2 \cdot 1627$, $3661 = 7 \cdot 523$), a wrong shape line ($\equiv 12 \pmod{16}$; correct $\equiv 4$), a wrong wave ordinal ("sixth"; correct "eighth") and a wrong 14592 ship time. The verification result itself (results.json, $\max|HH^T - nI| = 0$, 900/900 PASS) was unaffected; the verify code used the correct arithmetic.